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USEFUL PAPER MACHINE FORMULAE

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STOCK PREPARATION:

Conversion Factors for Viscosities:-

Pa-s	P	cP	lb/ft-s	lb/ft-h
1	10	1000	0.672	2420
0.1	1	100	0.0672	242
10^{-3}	0.01	1	$6.72*10^{-4}$	2.42

Mass Velocities:-

$V\rho = m / S \equiv G$

V = average velocity, m/s

ρ = density, kg/m^3

m = mass flow rate, kg/s

S = cross-sectional area of channel, m^2

G = mass velocity, kg/m^2-s

Fluid Velocities in Pipe:

Fluid	Type of Flow	Velocity Range Ft/s	m/s
Thin liquid	Gravity flow	0.5-1	0.15-0.30
	Pump inlet	1-3	0.3-0.9
	Pump discharge	4-10	1.2-3
	Process line	4-8	1.2-2.4
Viscous liquid	Pump inlet	0.2-0.5	0.06-0.15
	Pump discharge	0.5-2	0.15-0.6
Steam		30-50	9-15
Air or gas		30-100	9-30

Suction Lift and Cavitations :-

$NPSH = g_c / g\ [\{ (p_{a'} - p_v)/ \rho \} - h_{fs}] - Z_a$

NPSH = net positive suction head, m

g_c = newton's-law proportionality factor, 32.174 ft-lb/lb_f^{-s2}

g = gravitational acceleration, m/s^2 or ft/s^2

$p_{a'}$ = absolute pressure at surface of reservoir, atm or lb_f/ft^2

p_v = vapour pressure, atm or lb_f/ft^2

ρ = density, kg/m^3 or lb/ft^3

h_{fs} = friction in suction line, J/kg or ft-lb_f/lb

Z_a = height above datum plane at station a, m or ft

The velocity head at the pump inlet $\alpha_a\ \overline{V}_a^2/2g_c$ could be subtracted from the result given above to give more theoretically correct value of the available NPSH, but this term is usually only about 30 to 60 cm.

Area Meters : (ROTAMETERS)

In the orifice, nozzle, or venturi, the variation of flow rate through a constant area generates a variable pressure drop, which is related to the flow rate. The most important area meter is the rotameter. it consists essentially of a gradually tapered glass tube mounted vertically in a frame with the large end up. The fluid flows upward through the tapered tube and suspends freely a float (which actually does not float but is completely submerged in the fluid).

Theory and Calibration of Rotameter-

$F_D g_c = v_f\ \rho_f\ g - v_f\ \rho\ g$

F_D = drag force,N or lb_f

g = acceleration of gravity, m/s^2 or ft/s^2

g_c = Newton's law proportionality factor, 32.174 ft-lb/lb_f^{-s2}

v_f = volume of float, m^3 or lb^3

ρ_f = density of float, kg/m^3 or lb/ft^3

ρ = density of fluid, kg/m^3 or lb/ft^3

Stratified Blending in Storage Tank -

For effective blending in a large tank a side-entering propeller must be oriented precisely with regard to both its angle with the horizontal (for top-to-bottom circulation) and, in the horizontal plane, the angle it makes with the tangent to the tank wall at the point of entry. For optimum results this angle has been found to be between **80^0** and **83^0**.

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Jet Mixers-

Circulation in large vessels may also be induced by one or more jets of liquid. Sometimes jets are set in clusters at several locations in the tank. The behavior of a circular liquid jet issuing from a nozzle and flowing at high velocity into a stagnant pool of the same liquid. The velocity in the jet issuing from the nozzle is uniform and constant. It remains so in a core, the area of which decreases with distance from the nozzle. The core is surrounded by an expanding turbulent jet, in which the radial velocity decreases with distance from the centerline of the jet. The sinking core disappears at a distance from the nozzle of $4.3D_j$, where **D_j is the diameter of the nozzle**. The turbulent jet maintains its integrity well beyond the point at which the core has disappeared, but its velocity steadily decreases. The radial decrease in velocity in the jet is accompanied by a pressure increase in accordance with the Bernoulli principle. Fluid flows into the jet and is absorbed, accelerated, and blended into the augmented jet. This process is called **entrainment.** An equation applying over distances larger than $4.3D_j$ is –

$$q_e = (X/4.3D_j - 1)\, q_o$$

q_e = volume of liquid entrained per unit time at distance X from nozzle, m^3/s or ft^3/s
q_o = volume of liquid leaving jet nozzle per unit time, m^3/s or ft^3/s
X = distance from nozzle, m or ft

Uniform suspention of solid particles-

Zwietering's correlation is based on data for five types of impellers in six tanks from 6 in. to 2 ft in diameter. The critical stirrer speed is given by the dimensionless equation –

$$n_c D_a^{0.85} = S\, v^{0.1} D_p^{0.2} (g\Delta\rho/\rho)^{0.45}\, B^{0.13}$$

n_c = critical stirrer speed,r/s
D_a= agitator diameter, m or ft
S = shape factor,see table
v = kinematic viscosity, m^2/s or ft^2/s
D_p= average particle size, m or ft
g = gravitational acceleration, m/s^2
$\Delta\rho$= density difference, kg/m^3 or lb/ft^3
ρ = liquid density, kg/m^3 or lb/ft^3
B = 100 × weight of solid/weight of liquid

Shape factor S for critical stirrer speed –

Impeller type	D_t / D_a	D_t / E	S
6-blade turbine	2	4	4.1
D_a/W = 5	3	4	7.5
N_p =6.2	4	4	11.5
2-blade paddle	2	4	4.8
D_a/W =4	3	4	8
N_p = 2.5	4	4	12.5
3-blade propeller	3	4	6.5
N_p =0.5	4	4	8.5
	4	2.5	9.5

N_p = power number, $Pg_c/n^3D_a^5\rho$;
D_a = diameter of impeller, m or ft;
W = impeller width, m or ft.

Dispersion Operations-

In suspending solids, the size and the surface area of the solid particles exposed to the liquid are fixed, as is the total volume of suspended solids. In gas-liquid or liquid-liquid dispersion operations, by contrast, the size of the bubbles or drops and and the total interfacial area between the dispersed and continuous phases vary with conditions and degree of agitation. New area must constantly be created against the force of the interfacial tension. Drops and bubbles are continually coalescing and being re-dispersed.

$\psi = \pi N D_p^3/6$
ψ = volumetric fractional of dispersed phase, dimensionless
N = number of drops or bubbles
D_p= diameter of drop or bubble

Laminar Flow:-

At low velocities fluids tend to flow without lateral mixing, and adjacent layers slide past one another like playing cards.There are neither cross-currents nor eddies. This regime is called laminar flow. At higher velocities turbulence appears and eddies form, which lead to lateral mixing.

Viscosities of Gases and Liquids:-

$\mu / \mu_0 = (T / 273)^n$
μ = viscosity at absolute temperature T, K
μ_0 = viscosity at 0°C (273 K)
n = constant ≅ 0.65 for air, 0.9 for CO_2 and simple hydrocarbons, and 1.1 for SO_2 and steam.

Krofta save-all volume:-

Volume=$\pi * r^2 * h$

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Save-all Decision:-
Fiber loss, mt/year = (fiber in effluent, kg/min * 60 * 24 * 350) / 1000
Water with fiber in effluent, kg/min = fiber in effluent, kg/min * [(100 – felt pit cy.%) / (felt pit cy.%)]
Water with fiber in make-up water, kg/min = fiber in make-up water, kg/min * [(100 – felt pit cy.%) / (felt pit cy.%)]
Water with fiber in baled pulp, kg/min = fiber in baled pulp, kg/min * [(100 – baled o.d.fiber%) / (baled o.d.fiber%)]
Cost of lost fiber, Rs/year = (fiber loss, mt/year) * (Rs/mt of pulp)
Capital cost/savings = (save-all total installation cost, Rs) / (cost of lost fiber, Rs/year)
Maximum cost/savings ratio permitted = 2.0
If capital cost/savings ratio comes less than 2.0 then save-all must be installed.

Stock pump capacity:-
Pump throughput, kg/min = [pump capacity, l/min * cy.% of stock] / 100

Stock pump kWh:-
= discharge, kg/min * head, m * cy % * 0.0001635 = kWh

Stock pump kWh:-
= [(pump lifting, kg/min * 9.81 * lifting height, m) / 60000] * cy % = kWh

Reserve capacity for storage chests in between operation is 2 hours:-
Chest capacity each, m^3 = [(a.d. production, mt/hr*2*100) / (% consistency)] ÷ 5 chests

5 chests –
- raw chest
- refined chest
- mixing/blend chest
- m/c chest
- broke chest (3 times of chest capacity each)

Stock chest capacity:-
(M^3 * % Consistency) / 100 = Mt
1 ppm = 1 mg/ltr
1 ltr = 992 gm

Tank Sizing and Capacity:-
Tons = [#/ft^3 * volume, ft^3] / 2000
= [% B.D. * volume, ft^3] / 1.6 * 2000
#/ft^3 = weight of dry stock at % consistency
volume = volume of tank in ft^3
% B.D. = % consistency of stock

Water requirement for consistency$_1$ vs consistency$_2$:-
Water at 12 % cy for 1 mt/hr draw of o.d. pulp = 1mt * [(100 – 12) / 12] = 7.33 mt water / hr = 7.33 m^3/hr
Water at 4.5 % cy for 1 mt/hr draw of o.d. pulp = 1mt *[(100 – 4.5) / 4.5] = 21.22 mt water / hr = 21.22 m^3/hr
Water required in mt/hr from 12 % cy pulp to 4.5 % cy = 21.22 – 7.33 = 13.89 mt/hr flow rate = 13.89 m^3/hr

Refining theory:-
Specific Edge Load (SEL) –
SEL, watt-sec/meter = $[P – P_o] / L_s$
P = total power consumed by refiner when load is applied, kW
P_o = power consumed by refiner when load is not applied but pulp is running through, kW
L_s = cutting length, km/sec
Refining Capacity,(M) – [for DDR]
M, cm^3/sec = $[1/360]*\{[Z^2 * S_2^2 * n] / d_2^3\}*[d_1^3 – d_2^3]$
Z = number of bars
S_2 = width of bar knife at internal disc diameter d_2, mm
n = rotor rpm
d_1 = external diameter of disc, mm
d_2 = internal diameter of disc, mm [diameter of crushing zone]
Refining Capacity, (M) – [for Conical Refiner]
M, cm^3/sec = $\{[S_o * S_w * Z_w * n] / [120 * Cos\beta]\}*[(Z_a)l_2 + (Z_b)l_1]$
S_o = thickness of shell life, mm
S_w = thickness of rotor life, mm
Z_w = total number of bars in rotor
n = rotor rpm
β = angle of inclination of shell knives from horizontal
Z_a , Z_b = number of bars in two zones of shell
l_1 , l_2 = length of bars of rotor at longer and shorter, mm

Refining Surface of DDR, (F) –
F, cm^2 = $\{[d_1^2 – d_2^2] / [4\pi*d_2^2 * 10^2]\} * [S_2^2 * Z^2]$

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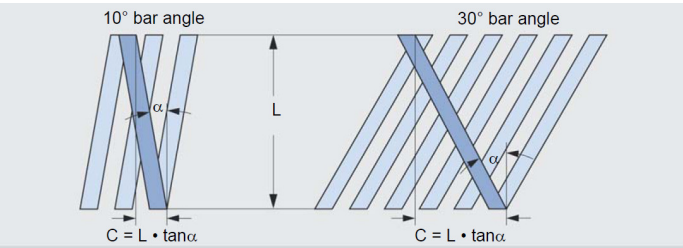
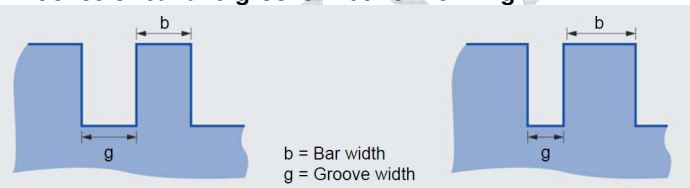
Numeral Hydration Factor of DDR, (R) –
R, km/cm^2 = [100 * L_s] / M
Cutting Capacity of DDR, (Z_{cut}) –
Z_{cut} , km/ton = L_s / [Q * 1000]
Q = designed capacity (ton/hr) of the refiner.
Power Consumption, (kW) – [voltage for P or P_o = 0.44 kV ; so, (0.671 * amp) = kWh]
[voltage for P or P_o = 3.3 kV ; so, (5.03 * amp) = kWh]
= √3 * voltage (kV) * current (amp) * power factor (Cos φ)
Cutting Length, (L_s) – [for **DDR**]
L_s , km/sec = I_c * I_o
I_c = intercuts/ sec = [n * Z^2] / 60
I_o = length of cuts of one stator or rotor knife calculation along the radius, km
Cutting Length, (L_s) – [for **Conical Refiner**]
L_s , km/sec = [n/60] * [Z_w /2] * [(Z_a) I_2 + (Z_b) I_1]
Definition of Cutting Edge Length
The cutting edge length CEL = Z_R × Z_S × I × (n/60)
Z_R = the number of rotor bars
Z_S = the number of stator bars
I = the length of the bars
n = the rotating speed
or
With a given set of fillings –
CEL = CLF. n
CLF = a fillings constant
n = the rotating speed
Parameters of the evaluation of refining
Specific edge load [J/m] = Net refining power [kW] / Cutting edge length [km/s]
Specific energy [kWh/t] = Net refining power [kW] / Throughput [t/h]
Influence of bar angle on number of intersection points –



$C = L \cdot \tan\alpha$ represents the bar crossing efficiency
$L_s \rightarrow L_{s,\gamma}$ $L_{s,\gamma} = L_{s,Stator} \cdot \tan\alpha_S + L_{s,Rotor} \cdot \tan\alpha_R$
$L_{s,\gamma} = L_s \cdot 2 \cdot \tan\alpha$ for identical design of rotor and stator
$SEL \rightarrow SEL_\gamma$ $SEL_\gamma = \frac{P_{net}}{L_S} \cdot \frac{1}{2 \cdot \tan\alpha} = SEL \frac{1}{2 \cdot \tan\alpha}$

Influence of bar and groove width on refining –



$\frac{b}{b+g}$ considers refining work of bar surface and probability of a fiber being treated
$L_{s,\gamma} \rightarrow L_{s,\gamma bg}$ $L_{s,\gamma = bg} L_{s,stator} \cdot \tan\alpha_S \frac{b_s}{b_s+g_s} + L_{s,Rotor} \cdot \tan\alpha_R \frac{b_R}{b_R+g_R}$
$L_{s,\gamma} = L_s \cdot 2 \cdot \tan\alpha \frac{b}{b+g}$ for identical design of rotor and stator
Modified **E**dge **L**oad ***MEL*** ***MEL*** = *SEL* • $\frac{1}{2 \cdot \tan\alpha} \cdot \frac{b+g}{b}$

Refiner fillings and their application –

Furnish	Hardwood pulps and secondary fibers		Softwood pulps	
Type of treatment	low intensity	standard	low intensity	standard
Spec. edge load SEL [J/m]	<1	<1.5 (1.7)	(1.5) 2 – 2.5	3 - 4
Cutting edge length CEL [km/s]	very high	high	average	low
Bar width [mm]	2	2 - 3	3	3 - 5
Groove width [mm]	(2) - 3	3 - 4	4 - 5	4 - 5
Cutting angle [°]	40 - 60	20 - 60	40 - 60	4 - 20

Tips –

- Specific edge load (SEL) gives an idea on the degree of intensity (refining action) in which the energy has been applied to the fibers.
- The degree of intensity with increasing edge load , the increasing of fiber treatment increases.
- Cutting angle is the angle resulting from interception of the bar angles, that is = sum of bar angles.
- Larger the cutting angle, the more fibrillation and more gentle the refining action.
- Edge length per second (L_s) is the total length of bar edges affective in a refiner within one second –
  $L_s = [n/60] * [Z^2] * \{[d_1 – d_2] / \cos \alpha \}$
  α = bar angle
- Chemical bamboo pulps are generally refined in the specific edge load ranges of **1. 0**, **1. 5**, **2. 0** watt-sec/meter.
- Chemical rice straw pulps are refined at a very low specific edge load ranges of **0. 275** to **0. 350** watt-sec/meter.
- DDRs are generally operated at low specific edge load between 1.5 and 2.5 watt-sec/meter.

WET END:

Reynolds Number and Transition from Laminar to Turbulent Flow:-

$N_{Re} = D\ \bar{V}\ \rho / \mu = D\ \bar{V} / v$
D = diameter of tube, m or ft
V = average velocity of liquid, m/s or ft/s
μ = viscosity of liquid,Pa-s or lb/ft-s
ρ = density of liquid, kg/m^3 or lb/ft^3
v = kinematic viscosity of liquid,μ/ρ, m^2/s or ft^2/s
N_{Re} = Reynolds number , in a pipe flow is always laminar at Reynolds numbers below 2100. Under ordinary conditions, the in a pipe or tube is turbulent at Reynolds numbers above about 4000.Between 2100 and 4000 a transition region is found where the flow may be either laminar or turbulent, depending upon conditions at the entrance of the tube and on the distance from the entrance.

Water per ton of pulp ratio (V):

Water per ton of pulp, V = [(100 – consistency,%) / consistency, %]

Press nip width:-

$2b = 2\sqrt{\Delta h} – 2\ Re$ = nip width
Δh = compression in nip, cm
b = half nip width, cm
$Re = (R_1 – R_2)/(R_1 + R_2)$
R_1 = roll radius, cm …..bigger
R_2 = roll radius, cm …..smaller

Internal water cooling for rubber covered press roll:-

General guide lines-

- when required nip pressure > 400 pli = 71.5 kg/cm; (1pli = 0.1786 kg/cm) and m/c speed > 600 m/min

then,

- Δtemp. Of cooling water should be < - 10^{0}F = < 5.5^{0}C
- flow rate should be sufficient to maintain metal/rubber below 70^{0}C

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Design of modern roll cover:-

	natural rubber	styrene butadiene	urethane
	20 – 30 P&J	20 – 30 P&J	20 – 30 P&J
water <65.5^{0}C	excellent	excellent	excellent
water >71^{0}C	ex	ex	fair
oils	fair	good	good
oil & water	fair	good	good
acids	ex	ex	fair
alkalies	ex	ex	fair

heat in { **conduction**, **radiation**, **transfer**, **hysteresis** } **heat out**

Hardness of roll covers can vary considerably depending on application:-

		P&J
1.	cylinder couch/lumpbreaker	200
2.	top press roll	0 or 15 – 20
3.	suction press roll	28 – 32
4.	bottom press roll plain	30 – 60
5.	wet felt rolls	0 – 5
6.	paper lead rolls	0 – 5
7.	breast roll	5 – 10
8.	table rolls	0 – 5
9.	wire guide roll	0 – 5
10.	contact pressing roll for drying	30 – 40
11.	size press hard roll	5 – 40
12.	size press soft roll	30 – 50
13.	MG pressure roll plain	15 – 20
14.	dryer felt rolls	5 – 10
15.	rotiformer couch plain	200
16.	rotiformer couch suction	120

0 – 50 P&J = hard
55 – 85 P&J = medium **P&J** 1/8″ ball
90 – 260 P&J = soft

Crown correction formula:-

$C = [N_2^2 - N_1^2]*[D_1 + D_2] / 2D_1*D_2$

C = additional crown required, i.e. the difference in diameter between the centre and 2″ in from the ends of the roll.
N_1 = nip width at centre of roll.
N_2 = nip width 2″ in from the ends of roll.
D_1 = diameter of top roll.
D_2 = diameter of bottom roll.

- if rolls have equal diameters, then :–

$C = [N_2^2 - N_1^2] / D$

D = diameter of roll (s) …..equal rolls

* the amount of total crown in a press is only good for one pair of rolls at one operating pressure.

Drainage Index (DI):-

$DI=b*c*v*10^{-3}*2.54$

b=constant for fabric CD support geometry
c=fabric count or strand count in MD
v=air permeability in CFM

e.g. *single layer fabric=13.1 *14 shaft double layer fabric=27.9
*double layer fabric=14.4 *triple layer fabric=27.4
*2½ layer fabric=28.2

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Fiber Support Index:-
FSI=⅔(aM+2bc)
a=constant for fabric MD support
b=constant for fabric CD support geometry
M=fabric mesh or strand count in CD
c=fabric count or strand count in MD
e.g. *single layer fabric=78 *14 shaft double layer fabric=136
*double layer fabric=97 *triple layer fabric=142
*2½ layer fabric=136

Flow of water:-
Q = A*V
Q=flow, m^3/sec
A=cross sectional area of pipe, m^2
V=average velocity of water, m/sec; V=14 m/sec at 1 kg/cm^2 pressure
$V=\sqrt{2gh}$ g=9.81m/sec^2; h=head of water column= w.r.t. 10 mWC=1 kg/cm^2 pressure;

Fluid velocity, fps:-
Velocity = [gpm * 0.321] / A
A = area, $inch^2$
Note : this formula is for save-all and general paper flow, since there is no orifice coefficient included.

Relationship between gallon per minute and pressure:-
$gpm_1 / gpm_2 = \sqrt{psi_1} / \sqrt{psi_2}$
1 gpm = 4.546 l/min

Felt pit cy.%:-
= [(fiber out of felt pit, kg/min) / (water out of felt pit, kg/min + fiber out of felt pit, kg/min)] * 100

Fiber out of press, kg/min:-
= [(water out of press, kg/min)*(cy.% of water leaving press)] / (100 – cy.% of water leaving press)

Fiber out of felt pit, kg/min:-
= (fiber leaving wire pit in overflow, kg/min) + (fiber out of press, kg/min)

Water out of felt pit, kg/min:-
= (water out of press, kg/min) + (water with fiber leaving wire pit in overflow, kg/min) + (felt showers, kg/min)

Water with fiber leaving wire pit in overflow, kg/min:-
= (fiber leaving wire pit in overflow, kg/min) * [(100 – cy.%) / cy.%]

Stock thickness on forming fabric:-
T = [BW] / [C*R* (J/W)]
T = thickness of stock on table, cm
BW = basis wt, gsm
C = consistency, %/100
R = retention from that point down the rest of the machine
J/W = jet to wire ratio = 1.0 except at slice
i.e. overall retention of a m/c with slice opening of ½″ making 50 gsm at 0.6% slurry and J/W ratio of 0.95
R = [0.0050] / [(0.0060) * (1.6) * (0.95)] = 73%

Fan pump power calculation:-
$P = [Q*\Delta p*(100/\eta)*K_3$
P = electrical power, kW
Q = flow rate, m^3/sec
Δp = pressure increase (suction to discharge), kPa = head
η = pump efficiency, %
K_3 = constant = [1/138.54]

Water in paper leaving press, kg/min:-
= production a.d., kg/min * [(100 – dryness after press %) / dryness after press %]

Water out of press, kg/min:-
= production a.d., kg/min * [{(100 – dryness before press%)/dryness before press%} – [{(100 – dryness after press%)/ dryness after press%]

Wire pit cy.%:-
= [(fiber to tray, kg/min) / (fiber to tray, kg/min + water to wire pit, kg/min)] * 100

Fiber balance on wire pit:-
= (fiber leaving wire pit in overflow, kg/min + fiber leaving wire pit in return to head box, kg/min) = fiber to tray, kg/min

Fiber balance on fan pump:-
= (fiber from stuff box, kg/min + fiber leaving wire pit in return to head box, kg/min) = fiber delivered to wire, kg/min

Head box cy.%:-
= (gsm) / (100*retention %*slice opening, m)

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Slurry from slice, kg/min:-
Wire speed, m/min*slice width, m*slice opening, m
Water to wire, kg/min:-
= (wire speed, m/min * wire width, m) * [(slice opening, m * 1000) – {(gsm / (10*retention %)}]
Fiber leaving wire in paper sheet, kg/min:-
= (wire speed, m/min * wire width, m * gsm) / 1000
Water leaving wire in paper sheet, kg/min:-
= [(wire speed, m/min * wire width, m * gsm)/1000] * [(100 – sheet cy.% before press)/sheet cy.% before press]
Water to tray, kg/min:-
= (water to wire, kg/min) – (water leaving wire in paper sheet, kg/min)
Fiber delivered to wire, kg/min:-
= [(wire speed, m/min * wire width, m * gsm) / (retention % * 10)] * 100
Fiber to tray, kg/min:-
= [{(wire speed, m/min * wire width, m * gsm) / (retention % * 10)} * 100] – [(wire speed, m/min * wire width, m * gsm) / 1000]
Overall fiber balance:-
Fiber out of felt pit, kg/min + fiber in baled pulp, kg/min = fiber from stuff box, kg/min + fiber in effluent, kg/min
Fiber balance at effluent:-
Fiber out of felt pit, kg/min = fiber in effluent, kg/min + fiber in make-up water, kg/min
Overall balance at stock chest:-
Fiber from stuff box, kg/min + water from stuff box, kg/min = [(fiber in baled pulp, kg/min*100)/baled fiber%] + [(fiber in make-up water, kg/min *100)/felt pit cy.%]
Formation – Blade Pulse Frequency:-
f, cycle/sec = [V, fpm] / 5 * λ, inches
V = wire speed, fpm
λ = blade spacing, tip to tip in inches
optimum frequency for formation improvement, f > 60 cycles/sec.
Head box volumetric flow rate:-
Head box volumetric flow rate, l/min = (70*P) / (B – A)
P=production, tpd
B=head box consistency, %
A=tray consistency, %
Head Box flow rate, gpm/inch:-
gpm/inch = S.O. * V * 0.052 * C
V = spouting velocity, fpm
S.O. = slice opening, inches
C = orifice coefficient; 0.95 for nozzle; 0.75 for low angle (converflow); 0.70 for high angle; 0.60 for straight (right angle)
Head Box approach system stock velocities:-
V, fps = [stock flow, gpm] * 0.0007092 / [pipe radius, ft^2]
= [stock flow, gpm] * 0.321 / [area of pipe, in^2]
acceptable range: 7 – 14 fps
Head box volumetric flow rate, ltr/min:-
= (wire speed, m/min * slice width, m * slice opening. m) * 1000[1 m^3 = 1000 ltr]
Basis wt , gsm:-
= [slice opening, mm * h/box cy % * 10]
Slice Opening, mm:-
= gsm / (10 * head box cy%)
Head box mass flow rate, kg/min:-
= (dry fiber rate, kg/min) / cy.%
Slice opening, mm:-
= (head box cy.% / gsm) * 1000
L/b ratio of slice geometry:-
L = bottom lip location (forward or backward distance corresponding to the upper lip tip), mm
b = slice opening, mm

- a lower L/b ratio around 0.5 means a steep jet angle into the wire and is frequently referred to as 'pressure forming'.
- a higher L/b ratio of 1.0 or greater means a relatively flat jet and is called 'velocity forming'.

Wire shake number:-
S = (f^2 * a) / m
S = shake number
f = frequency, shakes/min
a = amplitude, inch
m = machine speed, ft/min

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Drag load in fabric:-
= [volts*total amps*49] / [wire speed, m/min*fabric width, mm] kg/cm
Wear of forming fabric:-
% wear ={[C – A] / [0.58*C]} * 100
C = initial thickness strand
A = worn strand thickness
Fabric revolution:-
= [m/c speed, mpm / fabric length, m] * 1440 * days
One revolution:-
One revolution = 360^0 = 2 π radians
1 radian = 360^0 / 2 π = 57.3^0
Suction roll vacuum tension:-
$T_v = 10^{-3} * \mu * v * w$
T_v = max. tension difference due to vacuum, kg/m
μ = coefficient of friction (roll to wire), usually taken as 0.25 for metal to metal
v = vacuum level in suction roll, mmH_2O
w = width of suction box, mm

- when slipping occurs, the coefficient of friction, μ, may be only 40 – 60 % of the coefficient of creeping friction and slipping will continue until the wire/fabric stalls while the load on the drive can be quite small.

Suction couch vacuum:-
CFM = V*b*s*E*M
V = m/c speed, ft/min
b = roll shell face width, ft
s = hole depth, ft
E = % open area of shell
M = expansion factor, $(P_2/P_1)^{0.9} – 1$
P_2 = ambient pressure, inch Hg
P_1 = suction box vacuum, inch Hg
Liquid ring vacuum pump sealing water cascading system:-
From higher vacuum pumps (250 ~ 500 mm Hg) [Sealing water temp at 30 – 32 ^{0}C] pH=7 **to** pH=7 [sealing water temp at 27 – 43 ^{0}C] lower vacuum pumps (upto 250 mm Hg)
Shower oscillation speed calculation:-
Oscillation speed or traversing speed, mm/min = [m/c speed, mpm * jet, mm] / fabric length, m
Head of stock behind slice, mm Water Column:-
= $(\text{wire speed, m/min} / 265.7)^2 * 1000$
Total solids flow from head box, kg/hr:-
= [(total solids flow in paper, kg/hr) / FPR %] * 100
Standard Head box flow rate:-
gpm/inch =[B.D. ton / 24 hr / inch] * 16.76 * [1.5 – tray consistency, %] / 1.5 – net consistency, %
net consistency, % = h/box cy, % - tray cy, %
Tissue Head box flow rate:-
gpm/inch = [T.O. * V] / 19.25 = T.O. * V * 0.0052
T.O. = throat opening, inches
V = spouting velocity, fpm
Note : assume orifice coefficient = 1.0
Spouting velocity, fpm:-
$V = K\sqrt{h}$
V = spouting velocity, fpm
h = theoretical head
K = constant = 513.3 for inch of Hg head
= 481.5 for ft of H_2O head
= 139.2 for inch of H_2O head
= 732.3 for psig head
Head box discharge to wire, ltr/min:-
= [(total solids flow from head box, kg/hr) / (head box cy.%)] * 100 * (1/60)
Consistency, K, %:-
= [(wt. of dry material, T, kg) / (wt. of suspension, Q, kg)] * 100 [K, %=(T/Q)*100]
Flat box vacuum pump requirement:-
Water removed per kg of stock, liter/kg = 100 * [(1/incoming cy.%) – (1/outgoing cy.%)]
Amount of air removed per kg of paper, liter/kg:-
= (Water removed per kg of stock, liter/kg) * 10 [air:water = 10:1]

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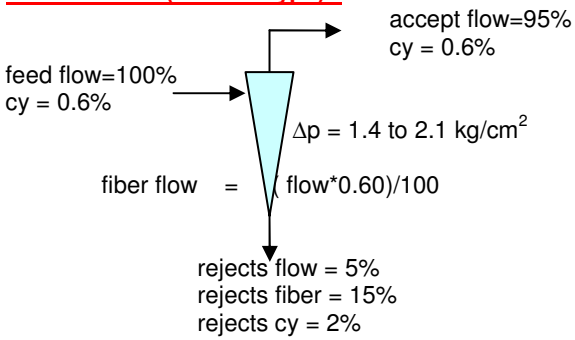
Amount of air per m^2 paper, liter/m^2:-
= [(Water removed per kg of stock, liter/kg) * 10 * gsm] / 1000
Amount of air per minute, m^3/min:-
= [{(Water removed per kg of stock, liter/kg)*10}*{(wire speed, m/min*wire width, m*gsm) / (10*retention%)}] /1000
this is the required air flow for the flat box and the vacuum pump of this capacity should be installed.
Head box energy balance:-
Kinetic energy = Potential energy
$\frac{1}{2} mv^2 = mgh$
$v^2 = 2gh$
$v = \sqrt{2gh}$
actually, $v = C_q \sqrt{2gh}$; C_q = energy loss = 0.85 to 0.90 for a tapered slice ;
or C_q = energy loss = 0.65 to 0.75 for an abrupt opening.
Vacuum pump capacity:-
Torricelli relationship
Velocity of air
$V = 60\sqrt{2gh}$; V=velocity of air, ft/min ; g=constant gravity acceleration, 32.2 ft/sec^2 ; h=head of fluid flowing, ft
So that, $V = 484\sqrt{h}$
For 'h' in inch of water, $V = 4005\sqrt{h}$
For 'h' in inch of Hg, $V = 14750\sqrt{h}$
Q = A*V ; V=average velocity, ft/min ; Q=volume, ft^3/min ; A=equivalent passage area, ft^2
Centricleaner (forward type):-



Press felt configuration:- (a sample)

Design	:	laminated/batt-on-mesh/batt-on-base/weft-less	
Felt weight	:	1150/1250/1350/1450 gsm	
Face layers	:	3×20D Nylon	15D = 15 Denier = 42 microns = 9 gm/m^2 water removal
		2×15D Nylon	20D = 20 Denier = 48 microns = 6 gm/m^2 water removal
Back layer	:	2×24D Nylon	24D = 24 Denier = 52 microns = 3 gm/m^2 water removal
MD Yarn	:	Piled Monofilament Yarn	
CMD Yarn	:	Piled Monofilament	
Base Weave	:	5 & 1 Single Layer	
Weight	:	700 gsm (two bases of 350 gsm each)	
Air Permeability	:	45 cfm	
Felt Thickness	:	3.2 mm	
Recommendation:		This felt can be used in pick up position also	

Forming fabric configuration:- (a sample)

Type	:	2.5 L 8 shaft
Warp	:	0.17 mm PE
Weft paper side	:	0.20/0.12 mm PE
Weft wear side	:	0.25 PE/0.25 PA
Ends/inch	:	155
Picks/inch	:	132
Permeability	:	427 cfm
Caliper	:	0.79 mm
Gsm	:	430
FSI	:	121
DI	:	31
Stretch at 4 kg/cm, %:		0.34
Void volume	:	478 cc/m

- Finer top surface for papermaking for higher retention
- Coarser machine side for stability wear resistance and ease of cleaning

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Vacuum capacity per slot:-

Vacuum Capacity, m^3/min = [slot area, m^2] * [average vacuum factor, 700 m^3/min/m^2]

- vacuum factor ranges from 660 – 970 m^3/min/m^2 for all types of felt and press configuration.
- air velocity in pipe line separator should be between 18 and 20 m/sec and after the separator it should be 28 to 30 m/sec.

Air flow for felt dewatering/felt conditioning:-

$V^1 = [0.069\ (P_{sp})^{0.476}\ (t_d)^{0.110}\ (V^*)^{0.916}] / (M_{p1})^{0.628}$

$M_{p2} = [1.23\ (M_{p1})^{0.819}] / [(V^1)^{0.024}\ (P_{sp})^{0.124}\ (t_d)^{0.096}$

V^1 = sp. Air flow through felt at suction tube, scfm/m^2

P_{sp} = pressure drop across felt at suction tube, inch Hg

t_d = dwell time of a particle of felt at suction tube, milliseconds

V^* = felt permeability, cfm/ft^2 at 0.5 inch H_2O

M_{p1} = felt moisture content approaching suction tube, lb H_2O/lb of felt

M_{p2} = felt moisture content leaving suction tube, lb H_2O/lb of felt

Actual air volume (CFM) at felt suction tube:-

= {[29.92 – operating vacuum at vac pump, inch Hg] / [29.92 – operating vacuum at suction tube, inch Hg]} * required CFM

Press felt cfm ranges:-

- tissue : 10 – 25 cfm
- fine paper : 25 – 80 cfm
- newsprint : 25 – 80 cfm
- liner/corrugating : 60 – 120 cfm
- cylinder : 70 – 150 cfm

Sheet / felt contact points:-

CMD : 2800 – 3500 points/cm^2

MD : 4100 – 4500

Shower water per gram of felt during sheet run:-

- I press : 0.1 gm of water / gm of felt
- II press : 0.08 gm of water / gm of felt
- III press : 0.06 gm of water / gm of felt

Mesh and Count for formic fabric:-

Mesh : it is the number of MD strands per inch of width.

Count : it is the number of CD strands per inch of width.

Mesh is normally given first ;

e.g. - 75×50, means, 75 MD strands / inch (mesh) and 50 CD strands / inch (count).

*the finer the mesh, the finer the paper grade/quality.

Required Dwell Time over Uhle Box between 2 to 4 milliseconds for successful water removal:-

Dwell Time, milliseconds = [(slot width, mm) / (m/c speed, m/min)] * 60

Nip load formula:-

Nip Load, kg/cm = {[2*(π/4)*d^2*operating pressure, kg/cm^2] ± roll weight, kg} / L_N, cm

d = diameter of cylinder piston (pneumatic or hydraulic etc), cm

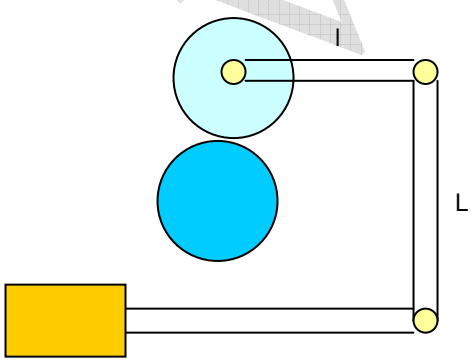
L_N = nip width, cm

2 = for both sides pressure (front + back)

- If the cylinder piston is connected with lever devices, then –

Nip Load, kg/cm = {[2*(π/4)*d^2*operating pressure, kg/cm^2* leverage, l/L] ± roll weight, kg} / L_N, cm

- Roll weight is added when loading is applied from top side to bottom side, and is subtracted when applied from Bottom side to top side.



Approximation for vacuum component in pli when taking nip impression, pli_v:-

pli_v = [vacuum box width, inches * vacuum level, inches of Hg] / 3

Roll speed, rpm:-

= [12 * fpm] / [π * D]
fpm = surface speed
D = diameter of roll, inches

Fabric life vs Drag load:-

Fabric life, days = $[1.23 * 10^6] * [\text{fabric tension, pli}]^{-2.55}$
1 pli * 0.175 = 1 kN/m

Felt tension calculations:-

For simple pulley system when sprocket pulleys are the same diameter and weight –
Tension, pli = total wt, lbs / [2 * clothing width, inches]

- If the weights are not marked, they can be weighed or calculated using 450 # /ft^3 or 0.26 # /in^3 or 7.2 gm/cm^3.

For sprocket pulleys having different diameters –
Tension, pli = [total wt, lbs * outer pulley dia, inch] / [2 * clothing width, inch * inner pulley dia,]

Method of measuring coefficient of friction, μ, for rubber covered roll:-

$F / W = e^{\mu A}$
F = tension in one side of the roll
W = tension in another side of the roll
A = contact area of roll's wrap angle
μ = coefficient of friction
e = base of Naperian logarithm

Calculation of press felt unit area weight:-

Z_t, gsm = [M*(58590)] / L*W
Z_t = unit area wt, gsm
M = press felt weight, lbs
L = press felt length, ft
W = press felt width, inches

Felt contaminations:-

- **Fiber and filler –**
  Alum, clay, TiO_2, cellulose, fines, pickout materials.
- **Organic materials –**
  Pitch, tar, oil, grease, asphalt, waxes, plastics, hot melts, stickies.
- **Solvent and heat set coatings –**
  Latexes, styrene, butadiene rubber latex, polyvinyl acetate.
- **Sizing agents –**
  Rosin, alkenyl succinic anhydride (ASA), alkyl ketene dimmer (AKD), wet strength resin.

Suction couch wasted volume:-

Wv = DA*(W)*U*(t)*ΔP/P
Wv = wasted volume, m^3/sec
DA = drilled area, %
W = drilled width, m
U = machine speed, m/min
t = shell thickness, cm
ΔP = vacuum at suction couch, mmHg
P = 760 mmHg vacuum

Forming Length guidelines:-

Dwell time in seconds between head box slice and flat box or dandy roll –

- Wire speed < 1200 fpm : 1.5 – 2.0 seconds
  Multiply forming length in ft by 40 (1.5 seconds) or 30 (2.0 seconds) to determine m/c speed that can be supported with conventional drainage table.
- Wire speed > 1200 fpm : 1.0 second
  Multiply forming length by 60 seconds to obtain m/c speed potential.
- 42 lbs liner : 1.25 seconds
  Multiply forming length by 48 to obtain m/c speed potential.
- Foodboard : 2.0 seconds
  Multiply forming length by 30 seconds to obtain m/c speed potential.

Wire Length, meter

Wire Length, meter = [(2 * Distance Centre Breast Roll to Centre Couch, mm) + (π/2) * (Breast Roll Dia, mm + Couch Roll Dia, mm) + (130 mm a constant for wire section)]/1000

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Vacuum Pump Selection Factors:-

Flat Box:		**Specific Required Air Flow =** 22.3 m^3/min/m wire width at 250 mmHg.
Suction Couch:	Single Box	**Specific Required Air Flow =** 220 $m^3/min/m^2$ suction box area at 500 mmHg.
Suction Pick Up:	Pick up Zone	**Specific Required Air Flow =** 220 $m^3/min/m^2$ suction box area at 500 mmHg.
	Holding Zone	**Specific Required Air Flow** = 22 $m^3/min/m^2$ suction box area at 250 mmHg.
	Nip Zone	**Specific Required Air Flow** = 241 $m^3/min/m^2$ suction box area at 500 mmHg.
Uhle Box:	Pick up Felt	**Specific Required Air Flow** = 660 $m^3/min/m^2$ suction box area at 250 mmHg.
	2nd Felt	**Specific Required Air Flow** = 530 $m^3/min/m^2$ suction box area at 250 mmHg.
	3rd Felt	**Specific Required Air Flow** = 430 $m^3/min/m^2$ suction box area at 250 mmHg.

Uhle Box Diameter Minimum, mm:-
√(air flow, m^3/hr * 19.89)
Uhle Box to Pre-Separator Minimum Diameter, mm:-
√(air flow, m^3/hr * 19.89)
Pre-Separator to Vacuum Pump Diameter, Minimum, mm:-
√(air flow, m^3/hr * 12.53)
Separator Diameter, mm:-
√(air flow, m^3/hr * 92.83)
Separator Height, mm:-
2 * separator diameter, mm
Barometric Leg Minimum Diameter, mm:-
√(air flow, m^3/hr * 8.7)
Barometric Leg Length Minimum, meter:-
(0.0136 m * vac reading, mmHg) + 0.9 m
Total Head:-
$h = S^2 / 527090$
S = jet speed in ft/min
h = total head in lb/in^2
Lip Opening:-
$H = [G.C_m/52] * [S_w/S_j] * [1/C_n]$
C_H = h/box cy%
R = wire retention %
C_n = net cy% = $C_H - C_w$
C_w = white water cy% = [1 – R] * C_H
C_m = reel dryness = [100 – reel moisture%] / 100
G = basis wt = lb / 1000 ft^2
S_w = wire speed, ft/min
S_j = jet speed, ft/min
H = lip opening, inch

DRY END:

Dimensional Equations :-
$q / A = 0.50[\Delta T^{1.25} / (D'_0)^{0.25}]$
q = rate of heat loss, Btu/h
A = area of pipe surface, ft^2
ΔT = excess of temperature of pipe wall over that of ambient (surrounding atmosphere), ^0F
D'_0 = outside diameter of pipe, inch.
Heat- Transfer Coefficient:-
$H_i = 0.023 G^{0.8} k^{0.67} c_p^{0.33} D^{-0.2} \mu^{-0.47}$
H_i = heat transfer coefficient
k = thermal conductivity
C_p = specific heat
D = diameter
μ = absolute viscosity
Kg_f:-
Kg_f = mass * gravity = 1 kg * 9.81 m/sec^2 = 9.81 Newton

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Heat Transfer by Conduction:-

Fourier's Law:-

The basic relation of heat flow by conduction is the proportionality between the rate of heat flow across an isothermal surface and the temperature gradient at the surface, at any location in a body and at any time.

$dq/dA = -k(\partial T/\partial n)$

A = area of isothermal surface,ft^2

n = distance measured normally to surface (⊥),ft

q = rate of heat flow across surface in direction normal to surface, Btu/h

T = temperature, ^0F

k = proportionality constant (thermal conductivity, Btu-ft-h-^0F)

Heat flow through a cylinder:-

$q = -k\ (dT/dr)\ 2\pi rL$

$2\pi rL$ = area perpendicular to the heat flow

r = radius of the cylinder

Evaporative Drying Rate Curve for Paper Machine:-

Y=Lb water/hr-ft^2

X=T_{sat} ^0F

- Tissue, average
  Y=(1.95/100)X-3.395
- Tissue, good
  Y=(2.125/100)X-3.0625
- Kraft, average
  Y=(203/100)X-3.83
- Kraft, good
  Y=(2.3/100)X-3.23
- Newsprint, average
  Y=(3.0/100)X-5.1
- Newsprint, good
  Y=(3.0/100)X-4.8
- Writing/Printing, average
  Y=(1.5625/100)X-2.40625
- Writing/Printing, good
  Y=(1.375/100)X-1.5375
- Paper Board, average
  Y=(1.40625/100)X-1.878
- Paper Board, good
  Y=(1.5/100)X-1.6
- Pulp, average
  Y=(1.32/100)X-2.27
- Pulp, good
  Y=(0.9375/100)X-0.96875
- Book Paper, average
  Y=(1.04166/100)X-0.93748
- Book Paper, good
  Y=(1.67/100)X-2.26
- Glassine&Greaseproof, average
  Y=(3.33/100)X-6.5
- Glassine&Greaseproof, good
  Y=(3.5/100)X-6.45

Specific Heat Capacity of water = 1.0 kcal/kg water-^0C

Specific Heat Capacity of steam = 0.46 kcal/kg steam-^0C at constant pressure.

Specific heat of paper web = 0.33 kcal/kg paper

Newton/meter = N/m = 0.001 kg/cm

1Btu = 0.252 kcal

1Btu/lb = 0.5556 kcal/kg

1Btu/ft^2-hr-^0F = 4.886 kcal/m^2-hr-^0C

1 kJ/kg = 2.326 Btu/lb , heating value

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"L" factor (lbs paper/ft^2 dryer surface/hr:-

"L" factor = SW / [("C" value) N]
L = # paper/ft^2 dryer surface/hr
S = m/c speed, fpm
W = basis weight, lbs/3000ft^2
N = number of dyers
"C" values
4 ft = 628.3
5 ft = 785.4
6 ft = 942.5

Drying Capacity:-

Drying Capacity, kg/hr = (total wrapped paper area)*(drying rate)
total wrapped area, m^2 = (N π *D*W*α)/360
N=number of drying cylinder
π=3.14
D=diameter of drying cylinder
W=deckle of paper
α=wrap angle=220^0
*drying rate = minimum=14kg/hr-m^2
maximum=30kg/hr-m^2
*Paper can be dried, kg/hr = (drying capacity)/M
M=kg of water /kg paper= (L/E) – 1
L=dryness leaving at pope reel
E=dryness entering dryer section

Paper m/c drying rate (R_w):-

R_w = (60*S*B*M) / N*A*π*D
R_w =drying rate, amount of water evaporated, kg/hr-m^2
S=m/c speed, m/min
B=basis wt of sheet as it leaves dryer section as dried (wet basis), kg/m^2
M=wt of water evaporated per unit wt of paper as dried (wet basis), kg water/kg paper=(L/E) – 1
N=number of paper dryers which contact the sheet
A=area of standard ream, 1.0 m^2
π = 3.1416
D=diameter of dryer cylinder, m
L=% dryness (wet basis) of sheet leaving the last cylinder
E==% dryness (wet basis) of sheet entering the first cylinder

Calculations of steam and condensate header pipe sizes:-

for steam header –
d_i , meter = √[(4D_{sc}) / (π*V_s*ρ_s*3600)]
d_i = inside diameter of steam header pipe, meter
D_{sc} = maximum steam consumption in the given group of cylinders, kg/hr [we assume that it is 40% more than the average consumption]
V_s = velocity of steam in pipe, m/sec [main header = 30 – 40 m/sec ; distribution header = 20 – 25 m/sec]
ρ_s = density of steam, say 2.62 kg/m^3
for condensate header –
it is 50 – 60% of the calculated cross-sectional area of the steam header.

Blow-through pipe size:-

At 3.45 kg/cm^2 steam pressure = 32 mm dia.
(@ 0.42 m^3/kg specific volume of steam and 1219 m/min steam velocity)

Recommended steam flow velocity:-

Minimum= 1219 m/min
Maximum= 1829 m/min

Steam requirement:-

Steam requirement, mt/hr = Q*1.5
Q=quantity of water evaporated, mt water/hr = (o.d. production, mt/hr)*M
M=kg of water /kg paper= (L/E) – 1
L=dryness leaving at pope reel
E=dryness entering dryer section

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Area of steam pipe:-

Area of steam pipe, A, cm^2 = (Q*1.5*1000*Vg*167)/V
Q=quantity of water evaporated, mt water/hr = (o.d. production, mt/hr)*M
Vg=recommended specific volume of steam @ 3.45 kg/cm^2 = 0.42 m^3/kg steam
V=average velocity of steam, m/sec

Diameter of steam pipe:-

Diameter of steam pipe, d, cm = $\sqrt{(4*A)/\pi}$
π = 3.14
A= area of steam pipe, cm^2 = (Q*1.5*1000*Vg*167)/V
Q=quantity of water evaporated, mt water/hr = (o.d. production, mt/hr)*M
Vg=recommended specific volume of steam @ 3.45 kg/cm^2 = 0.42 m^3/kg steam
V=average velocity of steam, m/sec

Sensible Heat:-

Sensible Heat = M*C*ΔT
M=mass wt., kg
C=sp. heat, kcal/kg
ΔT=temp. rise, ^{0}C

Thermal heat balance equation at Heat Exchanger:-

WF*C*(T_o – T_i) = SF*(H – h)
WF = cold water flow into the heat exchanger
C = coefficient of heat exchanger
T_o = temp of hot water outlet
T_i = temp of cold water inlet
SF = steam flow to the heat exchanger
H = inlet steam enthalpy
h = outlet condensate enthalpy

- one of greatest source of energy loss on paper machine.
- many can be removed from operation – not necessary.

Heat transfer co-efficient from steam to metal wall:-

=7000 kcal/m^2-hr-^{0}C

Heat transfer co-efficient from metal wall to paper:-

=245 kcal/m^2-hr-^{0}C

Water entering dryer section, kg/hr:-

= (fiber, kg/hr + filler, kg/hr)*(moisture %/dryness %)

Water lost in dryer section, mt/hr:-

= (water entering, kg/hr – water, kg/hr) / 1000

Total water to be evaporated:-

Kg water evaporated/hr = [tpd*1000*M, kg water/kg fiber] / 24

Approximate number of dryers required:-

N = [P*M] / [EV*T*D]
N = number of dryers required
P = production, kg/hr
M = kg water/kg fiber
EV = drying rate or evaporation rate, kg water/hr/ft^2
T = trim or deckle at reel, ft
D = dryer diameter, ft

Water with supply air, kg/hr:-

= [(air supply to hood, m^3/min) * (kg water/kg dry air at ^{0}C and % RH) * 60] / 0.8709 m^3/kg dry air

Sheet moisture removed from dryers, kg water/kg paper:-

= moisture leaving % - moisture entering %

Water removed from paper, kg/hr:-

= [(sheet moisture to dryer, kg water/kg paper – sheet moisture removed from dryer, kg water/kg paper)] * [(production, kg/hr)/(1 + sheet moisture removed from paper, kg water/kg paper)]

Heat to remove water, kJ/kg water:-

= [(steam flow to dryers, kg/hr) * (latent heat at saturated steam pressure, kJ/kg water)] / (water removed from paper, kg/hr)

Sheet moisture to paper, kg water/kg paper:-

= (dryness leaving % / dryness entering %) – 1

Sheet tension at calender:-

Tension, pli = (0.013 * basis wt, lbs/3000 ft^2)$^{0.3}$

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Water removal (Pressing vs Drying):-(a sample)
1 % change in sheet dryness at 40 % b.d.
M = [L/E] – 1
L = 95 %
E = 40 %
M = [95/40] – 1 = 1.375 kg water / kg paper at 40 % dryness after last press
Now,
M = [95/41] – 1 = 1.317 kg water / kg paper at 41 % dryness after last press
So that,
the differential water per kg paper will be dewatered = 1.375 – 1.317 = 0.058 kg water / kg paper
That is,
0.058 kg water is removed extra by press after increasing 1 % dryness of sheet
hence,
approximate reduction % in evaporative drying = (0.058 * 100) / 1.375 = **4.2** % (savings of operating steam cost)
In fact,
Sheet moisture ratio at 40 % dryness is 60/40 = 1.5
Sheet moisture ratio at 41 % dryness is 59/41 = 1.44
So,
The real difference is (1.50 – 1.44) = 0.06
Or (0.06 * 100) / 1.50 = 4.0 % [lower water content] steam drying saving when +1 % change in dryness from last press

Air permeability of dryer cloth:-

▪ Cotton felt	:	0 – 2 cfm/ft^2	:	0.33 m/hr
▪ Needle fabric	:	40 – 100	:	650 – 1630
▪ Open-mesh dryer fabric	:	30 – 400	:	490 - 6500
▪ Monofilament dryer fabric	:	30 – 1000	:	490 – 16300

*cfm is changed by typically weaving patterns.
**with 'stuffer' – low cfm ; without 'stuffer' – high cfm ;

Dryer fabric repairing:-
Excellent Idea-

- To reinforce the edge with sewing when the fabric is new, long before degradation and wear occur.
- The best commonly used stitch is the 'baseball' stitch.
- Best material to use these repairs – 'Aramid multifilament thread' for stitching.
- Frequently, a heat resistant adhesive is used to fix the sewing and keep it in place.
- It is to remember that the sheet must tolerate the extra pressure caused by sewing thread in the area of repair, otherwise, it is best simply to trim off loose threads and leave the tear un-repaired (usually when the tear or cut is primarily in MD).

Heat Energy Balance:-
Production, mt/hr = [speed, m/min] * [gsm] * [deckle, m] * [0.00006]
Fiber in sheet, mt/hr = Production, mt/hr * [dryness% /100]
Water in sheet, mt/hr = Production, mt/hr * [moisture% /100]
Coating solids, mt/hr = [speed, m/min] * [coating pick up, gsm]
Water in coating slurry, mt/hr = [coating solids, mt/hr] * [(100 – concentration, %) / concentration,%]
Total water brought in, mt/hr = [(water in coating slurry, mt/hr) + (water in sheet, mt/hr)]
Heat required at moisture, (x%):-
Water required at x% moisture, mt/hr = [(fiber in sheet, mt/hr) + (coating solids, mt/hr)] * [x% / (100 – x%)]
Sensible heat, fiber, kcal-^{o}c/hr = (fiber in sheet, kg/hr) * (specific heat of fiber, 0.33 kcal/kg) * (100 – sheet temp, ^{o}c)
Sensible heat, pigment, kcal-^{o}c/hr = (coating solids, kg/hr) * (specific heat of pigment, 0.2 kcal/kg) * (100 – sheet temp, ^{o}c)
Sensible heat, water, kcal-^{o}c/hr = (total water brought in, kg/hr) * (specific heat of water, 1.0 kcal/kg) * (100 – sheet temp, ^{o}c)
Latent heat, water, kcal-^{o}c/hr = (total water brought in, kg/hr) – (water required at x% moisture, kg/hr) * 540 kcal-^{o}c/kg
Total heat load, kcal-^{o}c/hr = (sensible heat, fiber, kcal-^{o}c/hr) + (sensible heat, pigment, kcal-^{o}c/hr) +
(sensible heat, water, kcal-^{o}c/hr) + (latent heat, water, kcal-^{o}c/hr)
Dryer heat output, kcal-^{o}c/hr = (Total heat load, kcal-^{o}c/hr) * (100 / dyer efficiency, %)

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HEAT ENERGY:

Steam Condensing Rate/Steam Consumption for Air:-
Rate of Condensation, Steam Consumption (kg/s)= [volumetric air flow rate, m^3/s]* [temperature difference in air flow, ^oC]*[specific enthalpy of air, $kJ/m^3\ ^oC$] / [specific enthalpy of evaporating steam, kJ/kg]

Efficiency of a heat machine:-
The efficiency of a heat machine working between two energy levels is defined in terms of absolute temperature:
$\eta = (T_h - T_c) / T_h = 1 - T_c / T_h$(1)
Where,
η = efficiency
T_h = temperature high level (K)
T_c = temperature low level (K)

FINISHING:

DIN format sizes of paper :- in mm [DIN = Deutsche Industrie Normen] ~ thought of the German Industry Standard
A series – [to be used for writing and printing papers]
4A0 = 1682*2378
2A0 = 1189*1682
A0 = 841*1189
A1 = 594*841
A2 = 420*594
A3 = 297*420 $y=x\sqrt{2}$
A4 = 210*297
A5 = 148*210
A6 = 105*148 $\sqrt{2}$= 1.414213562
A7 = 74*105
A8 = 52*74
A9 = 37*52
A10 = 26*37
A11 = 18*26 tolerance:- up to 150mm ± 1.5mm
A12 = 13*18 above 150mm ± 2.0mm
B series –
B_0 = 1000*1414 [to be used for envelopes, file folders, and so on]
C series –
C_0 = 917*1297 [to be used for envelopes, file folders, and so on]

Basis wt by sheet area:-
1 lb/1000 ft^2 = 4.8824 gsm [paperboard, liner]
1 lb/3000 ft^2 = 1.6275 gsm [ISO, newsprint]
1 lb/3300 ft^2 = 1.48 gsm [offset]
1 lb/1300 ft^2 = 3.76 gsm [bond]

Substance in use of template:-
Gsm = 10000w/ab
w = wt. in gm of specimen per test piece wt.
a = length in cm of specimen per test piece wt.
b = width in cm of specimen per test piece wt.

Ream wt.:-
Wt. in kgs for 500 sheets = wcd/2ab
w = wt. in gm of specimen per test piece wt.
a = length in cm of specimen per test piece wt.
b = width in cm of specimen per test piece wt.
c = length in cm of sheet in ream
d = width in cm of sheet in ream

Moisture content:-
Moisture content in % by wt = 100(W-w)/W
W = original wt. of conditioned specimen before drying
w = wt. of specimen after drying

Overall efficiency(OAE):-
OAE % = [m/c availability%/100]*[operating days/365]*[operating hours/24]*100

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Average gsm:-
Area, m^2 = (**weight**,gm) ÷ (**gsm**, gm/m^2) …….individually
X = [$area_1$+$area_2$+$area_3$+…..]
Y = [$weight_1$+$weight_2$+$weight_3$+…..]
Average gsm = Y/X
To calculate days to make quality wise production:-
Days = [total mt to make / (draw per hr x 24)] x [1 + {(100 - m/c efficiency %)/ 100}]

GENERAL:

Per ton paper:-

	Wood based	Agro based	Waste paper based
Fiber raw material, ton	2.2 – 2.5	1.6 – 2.0	1.25 – 1.40
Power, kwh	1500 – 1700	1000 – 1200	800 – 1000
Steam, ton	11 – 14	5 – 6	4 – 5
Water, m^3	200 – 250	150 – 200	100 – 150

Production, mt/day:-
= (gsm*deckle,m*m/c speed, m/min*1.44) / 1000
Paper machine speed, m/min:-
= {[(mt/day*1000) / 86400] / 0. (0gsm)] / deckle, m} * 60
Paper web draw:-
Draw, % = [(S_F – S_I) * 100] / S_I
S_F = final speed, fpm
S_I = initial speed, fpm
Estimated net weight of paper in reel form:-
= {π * [$(r_1)^2$ – $(r_2)^2$] * [1/bulk] * deckle} / 1000
r_1 = outer radius of the roll (paper), cms
r_2 = outer radius of the reeling spool (empty spool), cms
bulk = cc/gm
deckle = cms
π = 3.14
Machine speed vs gsm:-
m/c $speed_1$ * gsm_1 = m/c $speed_2$ * gsm_2
Fiber, kg/hr:-
= production, mt/day*(1000/24)*[(dryness %)/100]*(100 – filler %)/100
Water, kg/hr:-
= production, mt/day*(1000/24)*(moisture %/100)
Filler, kg/hr:-
= production, mt/day*(1000/24)*[(dryness %)/100]*(filler %/100)
Tons per Day (TPD):-
TPD = [capacity, gpm * b.d. consistency%] / 16.65
Dry fiber rate:-
= basis wt * speed * width
1 grain = 64.799 mg
7000 grains = 1 lb
15432 grains = 1 kg
Breaking Length:-
BL (km) = (tensile strength kg/cm^2)/(sp. gravity of paper * 99.98 kg/cm^2/km)
Sheet temperature meter:-
Swema contact pyrometer sensor to measure sheet temperature
Or Infrared (IR) sensor
Steam showers over vacuum boxes at wet end:-
Devronizer steam shower to use for hot pressing
Power calculation at running load of a motor:-
P = √3 * V * I * Cos φ
P = power in watt
√3 = 1.732
V = volts
I = running load in amp
Cos φ = power factor, 0.8

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Dryer screen cleaner:-

TROSIREX from W&F

- NaOH = 2.0 %
- Mixture of Aromatic Hydrocarbons = 25.0 %

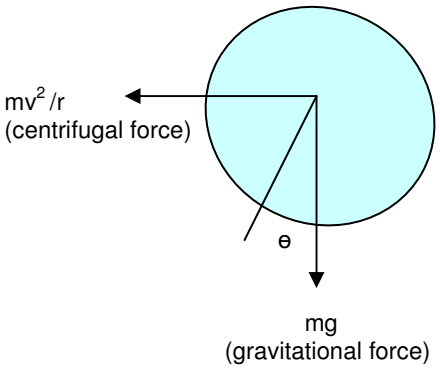
Roll grinding:- (gauge meter)

1 reading = 1 division = 0.01 mm
e.g. 47 reading = 47 divisions = 47 * 0.01 = 0.47 mm
1 thou = 1 inch /1000 = 25.4 mm /1000 = 0.0254 mm
so that, 0.47 mm = 18.5 thou [0.47/0.0254]

Strength Index or Paper Stength:-

$= [BF * TF * \log(\text{double folds})]^{1/2} * 100$

Dryer syphon angle vs m/c speed:-



m = mass
v = dryer surface speed, m/min
r = radius of dryer, m
g = gravity constant, 9.81 m/sec^2
ɵ = siphon angle corresponding to 'g'
so that,

$$\tan ɵ = (mg) / (mv^2/r) = g\, r / v^2$$
$$ɵ = \tan^{-1} [g\, r / v^2] = \text{syphon angle}$$

Doctor load, hp/inch/100 fpm:-

= (μ * pli) / 330
μ = 0.25 – 0.30 for metal ; μ = 0.17 – 0.25 for plastic ; μ = 0.22 – 0.27 for fiber glass ; = friction coefficient ;
pli = pounds per linear inch = lbs/inch

Critical Speed of calender roll, fpm:-

$C.S. = 4.12 * 10^6 * (R_o/L^2) * (\sqrt{R_o^2+R_i^2})$
C.S. = critical speed, fpm
R_o = outside radius, inches
R_i = inside radius, inches
L = centerline to centerline bearing, inches, [assume L = face + 40 inches]

Approximately Critical Speed of a roll:-

$C.S. = [55.37 * D_o * 0.9] / \sqrt{d_r}$
C.S. critical speed
D_o = outside diameter of roll, inches
d_r = roll deflection (inches) over face due to roll wt only (not to include externally applied forces)
$d_r = wF^3 * [12B – 7F] / 384 * E * I$
w = resultant unit load of sheet, lbs/in
F = shell face, inches
B = centerline to centerline bearing, inches
E = modulus of elasticity, lb/in^2
I = moment of inertia, inches^4 = $0.0491 * [D_o^4 – D_i^4]$
D_i = inside diameter, inches

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Estimated relative cost per ton of paper to dewater the sheet:-

- Forming fabric = 10 %
- Press section = 12 %
- Dryer section = 78 %

Paper machine central lubricating systems:-

Lube/oil cleanliness level ~ above 5000 particles (> 5μm) / ml [ISO Code 19/17]
Recommended oil cleanliness ~ at or below 160 particles (> 5μm) / ml [ISO Code 14/12]
with properly designed Silt-Control Filtration System
to achieve maximum bearing life.

Yield, % :-

= {[dry product mass out] / [dry material mass in]} * 100

Kappa number:-

Kappa number is the number of ml of 0.1 $KMnO_4$ consumed by 1 gm of pulp in 0.5 N H_2SO_4 after a 10 minute reaction time at 25 ^{0}C under condition such that ½ of the permanganate, M_n , remains unreacted.

- Klason lignin, % = 0.15 × kappa number

'K' number (permanganate number):-

Log [kappa number] = 0.837 + 0.0323 (40 ml 'K' number)

Roe number:-

Roe number = 0.158 × kappa – 0.2 [for kraft]
Roe number = 0.199 × kappa + 0.1 [for sulfite]

Chlorine number ['C', hypo number]:-

Chlorine number = 0.90 × Roe number

Coating pigment dispersion time:-

Dispersion time, min = [wt of clay, mt * 900] / [mixer motor rating, kWh * 0.7]

Horse Power:-

HP = TN / 63000
T = torque, inch – pounds
N = speed, fpm

Tension HP:-

Tension HP = [fpm * pli * inches of width] / 33000

Motor torque:-

Motor Torque, lb-ft = [HP, rdc * 5252] / motor speed in rpm

Synchronous speed of AC induction motor:-

Synchronous Speed, rpm = [rated frequency * 120] / number of poles

AC feeder transformer selection formula:-

KVA = (1.1) * (hp)
KVA = ac transformer KVA required by the drive = Kilo Volts Ampere
hp = selected drive horsepower

Total winder hp requirements:-

Winder hp = [web tension, lbs * web speed, ft/min * full roll diameter, inches] / [33000 * empty core diameter, inches]

Required winding tension:-

Winding Tension, lbs/inch = [basis wt, lbs/3000ft^2] / 20

Roll rpm:-

= [line speed, ft/min] / roll circumference, ft

Dandy Roll rpm:-

rpm = [wire speed, fpm] / [3.142 * dandy roll diameter, ft]
target = 125 – 150 rpm

Size Press Roll rpm:-

rpm = [web speed, fpm] / [3.14 * size press roll diameter, ft]
target = 250 rpm

Motor rpm:-

= [speed, ft/min * gear ratio] / [π * diameter of roll, ft]

Winding tension:-

Winding Tension, lbs = web tension, pli * web width, inches

V-notch flow rate:-

Q, ft^3/sec = (8/15) *C_d * (√2g) * tan (θ/2) * (H, ft) $^{5/2}$
C_d = coefficient of discharge = 0.6 for 90^0 V-notch
Empirical Formula –
Q, ft^3/sec = 2.5 H$^{2.5}$ [Q = flow rate, ft^3/sec; H = upstream head, ft]
ft^3/sec = 0.0283169 m^3/sec

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Flow Rate in pipe/Jet:-

a. low nappe (low discharge height)
   $Q, m^3/sec = 5.47 * D^{1.25} * H^{1.35}$ equation – 1
   Q = flow rate, m^3/sec
   D = diameter of pipe, m
   H = discharge height, m

b. jet
   $Q, m^3/sec = 3.15 * D^{1.99} * H^{0.53}$ equation – 2
   Q = flow rate, m^3/sec
   D = diameter of pipe, m
   H = discharge height, m

   If $H < 0.4 D$ use equation – 1
   If $H > 1.4 D$ use equation – 2
   If $0.4 D < H < 1.4 D$ calculate both equations and take the average.

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